Panskura Banamali College(Autonomous)

2nd Semester-2022

Paper Code: COSHC4T

Paper Name: Discrete Structures Theory

Full Marks: 60 Time :3 Hours

Answer any *four* questions: 4×15 = 60

1. i) Define cartesian product of two sets. Prove that $A \times (B - C) = (A \times B) - (A \times C)$, where A, B, C are three subsets of universal set S.

   ii) Give an example of relation which is both reflexive and transitive but neither symmetric nor antisymmetric. How many reflexive and symmetric relations are there on a set with n elements?

   iii) Let $f: A \to B$ and $g: B \to C$ be both invertible mappings. Then prove or disprove that $g \circ f: A \to C$ isinvertible and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. Is $(g \circ f)^{-1}$ a bijection ?

   (1+4)+(2+3)+(4+1)

2. i) Define power set of a set. Prove that the cardinality of the power set of a set having cardinality n is 2^n. Is power set of a set S a partition of S ?

   ii) State Pigeonhole principle. Using the principle of mathematical induction prove that $1.1! + 2.2! + 3.3! + \cdots + n.n! = (n+1)! - 1$, where n is a natural number.

   iii) Define closure of a relation. Find the reflexive and transitive closures of the relation R={(a, b),(b, c),(c, a),(d, b)} on the set A={a, b, c, d}. Is it possible to find a transitive closure of any binary relation?

   (1+3+1)+(1+4)+(1+3+1)

3. i) Define tautology. Construct the truth table for (~**P**→~**Q**)↔(P→Q). Is it tautology?

   ii) Suppose that $f(x)$is $O(g(x))$. Does it follow that $2^{f(x)}$is $O(2^{g(x)})$? Show that $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$ is $O(\log n)$.

   iii) Find the converse, contrapositive and inverse of the proposition *"I come to the class whenever there is a practical session."*

   (1+4+1)+(2+4)+3

4. i) Give a combinatorial proof of $\sum_{k=1}^{n} k\binom{n}{k}^2 = n\binom{2n-1}{n-1}^2$. Find the co-efficient of $x^2y^{1000}z^{1020}$ in the expansion of $(2x+4y-z)^{2022}$.

   ii) Define generating function for a sequence of real numbers. Find the generating function for the sequence {1, 2, 4, 8,.....} . Using generating functions or otherwise solve the recurrence relation $a_n = 7a_{n-1} - 16a_{n-2} + 12a_{n-3} + n4^n$ with initial conditions $a_0 = -2, a_1 = 0$ and $a_2 = 5$. (5+2)+(1+2+5)

5. i) State the principle of inclusion and exclusion for two sets. A computer company receives 350 applications from computer graduates for a job planning a line of new web servers. Suppose that 220 of these applicants majored in computer science, 147 majored in computer application and 51 majored in both. How many of these applicants majored neither in computer science nor in computer application?

   ii) What will be the number of components of a connected simple graph? Prove that a simple graph with n vertices and k-components can have at most $\frac{(n-k)(n-k+1)}{2}$ edges.

   iii) Is it possible to construct a graph with 13 vertices such that 5 vertices has degree 4 and remaining vertices of degree 4? What will be the number of edges?

   (1+4)+(1+5)+(3+1)

6. i) Prove that a connected multigraph has an Euler path but not an Euler circuit ifand only if it has exactly two vertices of odd degree.

   ii) What is the chromatic number ofK_n? Define complete bipartite graph. Prove or disprove that a complete bipartite graph with n vertices has $\left\lfloor \frac{n^2}{4} \right\rfloor$ edges.

   iii) Show that a circuit free graph with n vertices and n-1 edges is a tree.

   5+(2+1+3)+4

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